ANALYZING GROCERY STORE DATA USING ML

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1 INTRODUCTION

In this project, we will analyze the marketing campaign data of a grocery store for the sake of extracting useful information that will help increase the store's profitability. In general, marketing campaign data can reveal very actionable insights if we look at the right things. We will be employing a road map for practically analyzing the said data in Python using ML.

2 DATA PREPARATION

A well-known marketing campain data of a grocery store, namely `marketing_campaign.csv`, obtained from `#Github` was loaded and inspected using the appropriate python modules and techniques (see Figure 1). Further inspection and structure investigation were also done. The crucial data cleaning was also performed to remove null and/or NaN entries.

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```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2240 entries, 0 to 2239
Data columns (total 29 columns):
 #   Column               Non-Null Count  Dtype
---  ------               --------------  -----
 0   ID                   2240 non-null   int64
 1   Year_Birth           2240 non-null   int64
 2   Education            2240 non-null   object
 3   Marital_Status       2240 non-null   object
 4   Income               2216 non-null   float64
 5   Kidhome              2240 non-null   int64
 6   Teenhome             2240 non-null   int64
 7   Dt_Customer          2240 non-null   object
 8   Recency              2240 non-null   int64
 9   MntWines             2240 non-null   int64
 10  MntFruits            2240 non-null   int64
 11  MntMeatProducts      2240 non-null   int64
 12  MntFishProducts      2240 non-null   int64
 13  MntSweetProducts     2240 non-null   int64
 14  MntGoldProds         2240 non-null   int64
 15  NumDealsPurchases    2240 non-null   int64
 16  NumWebPurchases      2240 non-null   int64
 17  NumCatalogPurchases  2240 non-null   int64
 18  NumStorePurchases    2240 non-null   int64
 19  NumWebVisitsMonth    2240 non-null   int64
...
 27  Z_Revenue            2240 non-null   int64
 28  Response             2240 non-null   int64
dtypes: float64(1), int64(25), object(3)
memory usage: 507.6+ KB
```

Figure 1: Information of original data.

3 FEATURE ENGINEERING

Investigation of the categorical data was done and also some feature engineering like combining columns, creating new ones, renaming some columns and deleting repeated ones were also done. This is to prepare the data for ML techniques. Moreover, outliers were found and removed by setting a cap on the appropriate features. Additional feature engineering like encoding (using `LableEncoder()`) and scaling (using `StandardScaler()`) were done. The relative boxplot of the data features is depicted in Figure 2.

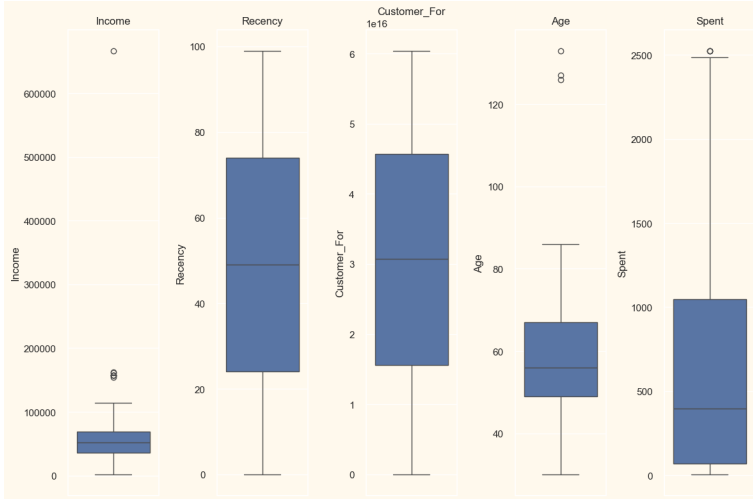


Figure 2: Relative boxplot of data features.

4 CUSTOMER SEGMENTATION

Since all customers are not equally profitable, we need to identify those who spend high in addition to those who visit the store frequently. After feature engineering, we were able to know the spending ability of the customers and identify high-spenders. By computing mean, median and max of the spendings, we were also able to realize the existence of strong spending imbalance. AS depicted in Figure 3, it is worth noticing that the top 30% help generate 70% of the total revenue. They are 665 (out of 2212) in number and we recommend to design personalized campaigns for them. Moreover,

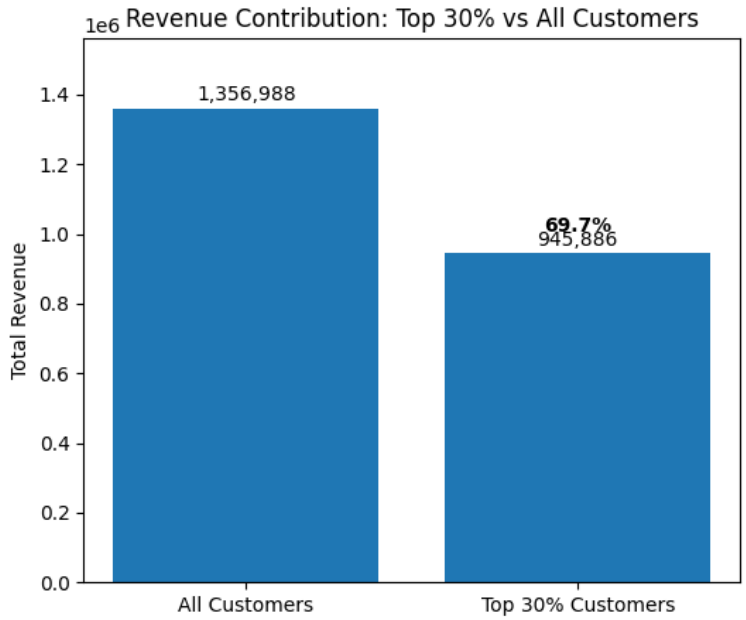
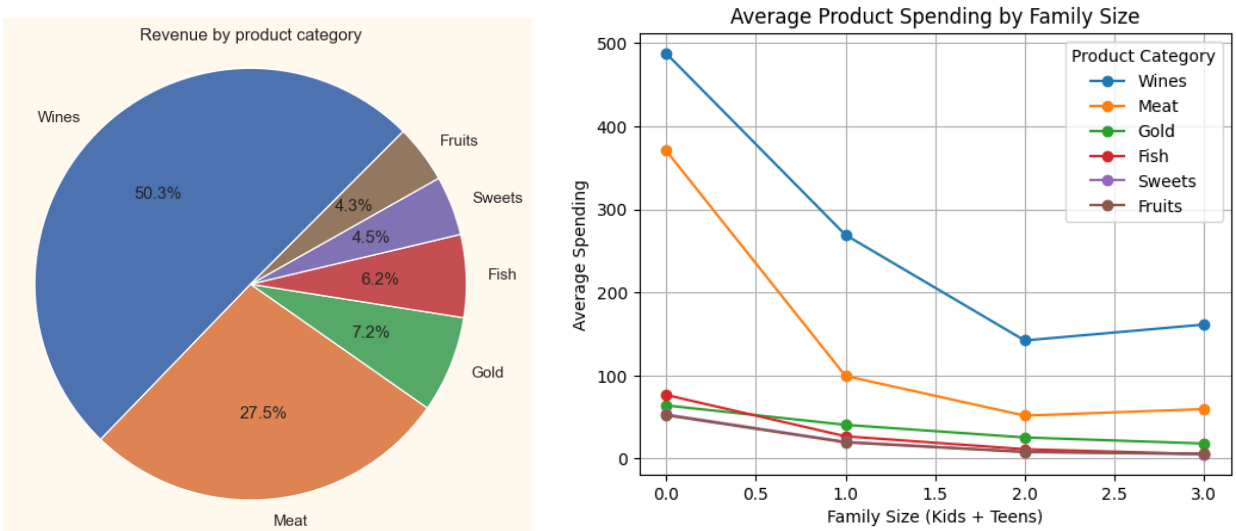


Figure 3: Revenue contribution.

the frequencies of the customers were calculated which helped as identify the regular visitors of the store. It is recommended to invest marketing budget on high-sepnders and frequent visitors.

5 PRODUCT CATEGORY PROFITABILITY

As depicted in Figure 4 (a), Wine and Meat are the core profit engines. Wine being the dominant revenue driver. These two are responsible for 50% and 28% of the total revenue, respectively. Therefore, it is recommended to feature wine and meat bundles in the marketing strategy. The analysis in this section tells us to promote wine and premium products for small families while bulk meat and food bundles for large families. Moreover, smaller families tend to have higher per-household spending whereas larger families spend more on necessities, less on premium items (Figure 4 (b)). These imply that it also helps to design family-size specific promotion strategies.



(a) (b)

Figure 4: Product profitability.

6 CAMPAIGN PERFORMANCE

Here we analyze how much the conducted campaign was effective. From Figure 5, it can be observed that some campaigns perform better than others but most customers don't accept any campaign (1754 out of 2212) implying the existence of a campaign fatigue. Therefore, it is wise to focus on customers who accepted at least one past campaign (only 136 out of 2212) and stop sending campaigns to chronic non-responders.

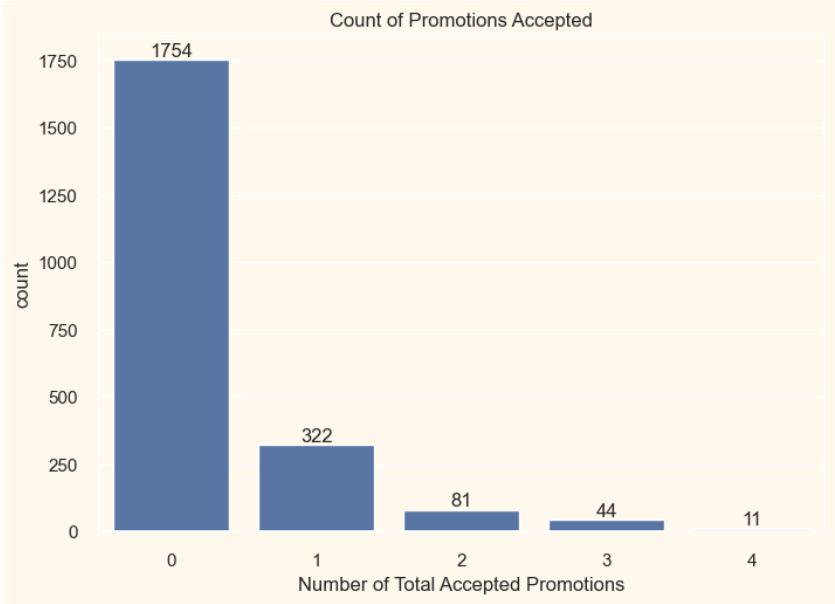


Figure 5: Campaign performance.

7 RFM ANALYSIS

The RFM analysis, depicted in Figure 6, helped us to categorize the customers as Champions or profit gold (with RFM 11-12), Loyal (RFM 9-10), At-risk (RFM 6-8) and Lost (RFM 5 or less). Therefore, by mapping the customers with their RFM scores, we might design VIP programs for customers with RFM 9 and higher, win-back campaign for those with RFM 6-8 and stop expensive marketing for RFM 5 and less.

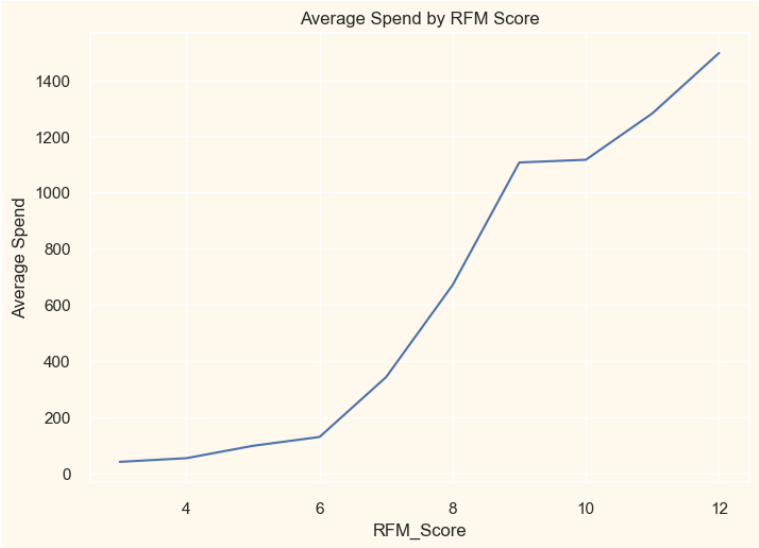
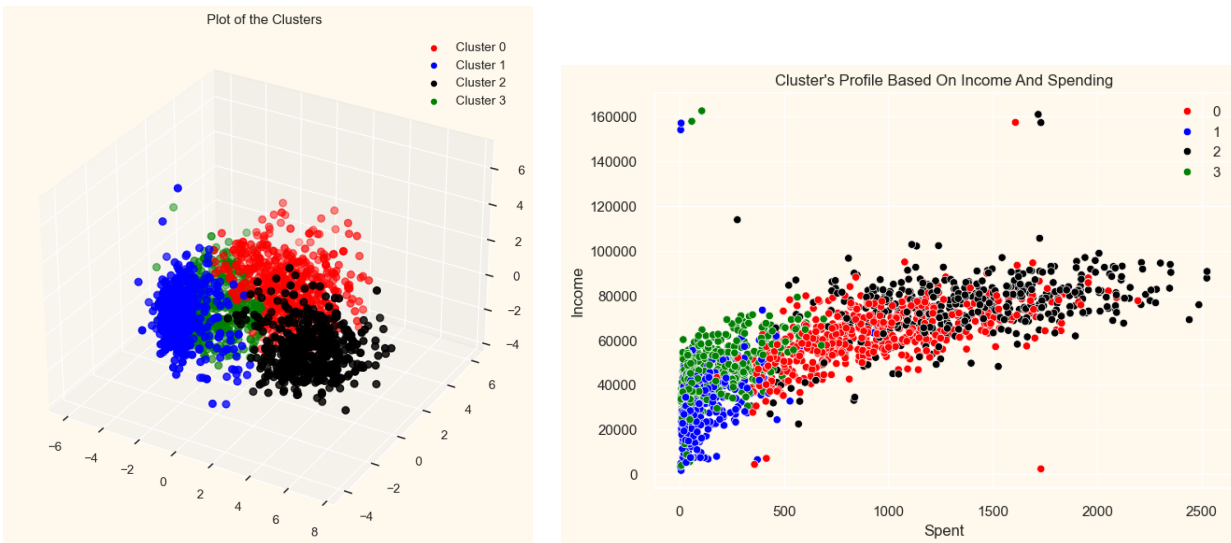


Figure 6: RFM analysis.

8 DIMENSIONALITY REDUCTION with PCA

First off, the correlation matrix heatmap was plotted for the sake of identifying highly correlated features. Thereafter, we used PCA in collaboration with Agglomerative Clustering and the Elbow method to calculate the optimal number of clusters, which we found to be 4 (Figure 7 (a)). From Figure 7 (b) it can be observed that the clusters were more or less balanced without any additional manipulation. Among the clusters, the most important ones that help the store to generate more profit were found to be Cluster 0 (those with middle income but spend high) and Cluster 2 (those having moderate income but are high spenders). Assuring that it is spending that matters in the store's profit, not just income!



(a) (b)

Figure 7: Dimensionality reduction with PCA.

9 PREDICTION MODELING AND FEATURE IMPORTANCE

Using `RandomForestClassifier`, we were able to identify the top three important features which are *Spent (Total Spending), Recency* and *Income* telling us that customers which purchase frequently and engage with the store on a regular basis (and spend more) are the ones that help generate more profit. See Figure 8 for more detail.

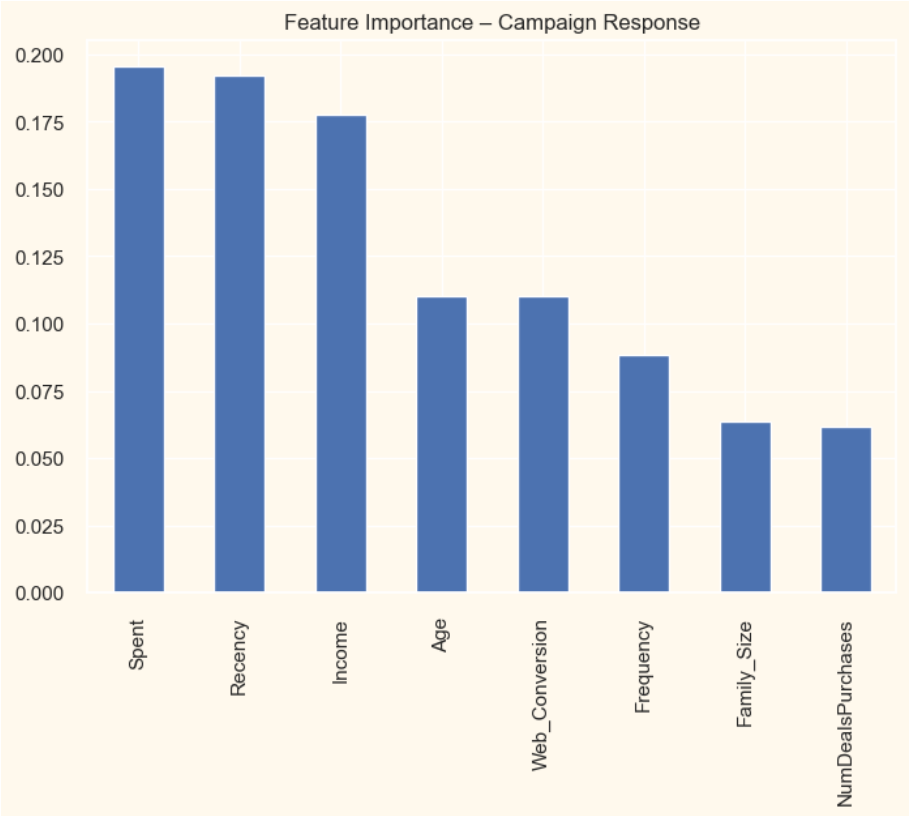


Figure 8: Feature importance.

10 BUSINESS INSIGHT

To ensure store profitability, it is highly recommended to:

- Focus on top spenders as they help generate 70% of the total revenue.
- Promote Wine and Meat aggressively.
- Design loyalty, personalized offers and/or VIP programs for customers with RFM ≥ 9 to retain them at all cost.
- Work on win-back campaigns for those with RFM 6-8 and stop expensive marketing on customers with RFM 5 and less.
- Allocate more campaiging budget for customers in Cluster 0 (those with middle income but spend high) and Cluster 2 (those having moderate income but are high spenders).
- Design promotion strategies that take into account the most important features (Spending, Recency and Income).
- Avoid campaigning to non-responders.
- Don't treat customers equally.